

Problem

Refer to the network in Fig. 16.98. Find the following transfer functions:

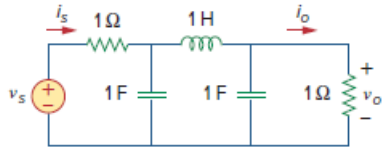
(a) $H1(s) = V_o(s)/V_s(s)$

(b) $H2(s) = V_o(s)/I_s(s)$

(c) $H3(s) = I_o(s)/I_s(s)$

(d) $H4(s) = I_o(s)/V_s(s)$

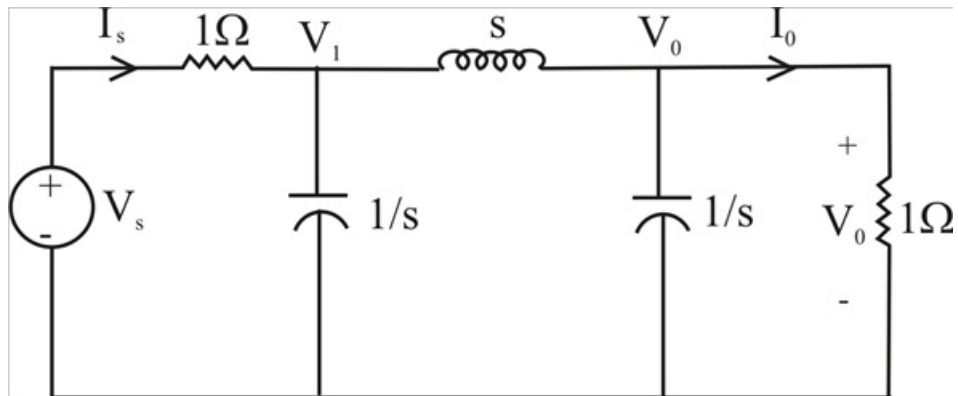
Figure 16.98:



Step-by-step solution

Step 1 of 6

(a) Consider the following circuit



Step 2 of 6

At Node V_1 ,

$$V_s - V_1 = sV_1 + \frac{V_1 - V_0}{s}$$

$$V_s = \left(1 + s + \frac{1}{s}\right)V_1 - \frac{1}{s}V_0 \dots \dots \dots (1)$$

At Node V_0 ,

$$\frac{V_1 - V_0}{s} = sV_0 + V_0 = (s+1)V_0$$

$$V_1 = (s^2 + s + 1)V_0 \dots \dots \dots (2)$$

Step 3 of 6

Substituting (2) into (1)

$$V_s = \left(s + 1 + \frac{1}{s}\right)(s^2 + s + 1)V_0 - \frac{1}{s}V_0$$

$$V_s = (s^3 + 2s^2 + 3s + 2)V_0$$

$$H_1(s) = \frac{V_0}{V_s} = \frac{1}{s^3 + 2s^2 + 3s + 2}$$

Step 4 of 6

(b) $I_s = V_s - V_1$

$$I_s = (s^3 + 2s^2 + 3s + 2)V_0 - (s^2 + s + 1)V_0$$

$$I_s = (s^3 + s^2 + 2s + 1)V_0$$

$$H_2(s) = \frac{V_0}{I_s} = \frac{1}{s^3 + s^2 + 2s + 1}$$

Step 5 of 6

(c) $I_0 = \frac{V_0}{1}$

Therefore $I_0 = V_0$

$$H_3(s) = \frac{I_0}{I_s} = \frac{V_0}{I_s} = H_2(s) = \frac{1}{s^3 + s^2 + 2s + 1}$$

Step 6 of 6

(d) Since $I_0 = V_0$

$$H_4(s) = \frac{I_0}{V_s} = \frac{V_0}{V_s} = H_1(s) = \frac{1}{s^3 + 2s^2 + 3s + 2}$$